

Ra-226 to Ac-225 Surrogate Study

What is already complete, what the frozen model can and cannot support, and the specific outside evidence now needed.

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PURPOSE

I am seeking public data, references, or a brief technical review to determine whether a frozen surrogate can support fast, solver-verified screening of Ac-225 production scenarios. I am not requesting private facility data, a radioactive experiment, clinical advice, or an endorsement.

0.97% Matched synthetic worst-mesh median Ac-225 endpoint error, seed 42	3.94% / 183.84% Median / p95 error across 9,000 productive cases in a 10,000-case grid	Rank 2 V3-selected schedule after Radau verification; 0.047% objective regret	84.9x / 56.7x Faster than Radau, but slower than the exact analytic batch baseline at 10,000 cases
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Current Technical Contract

Frozen model and pathway - Ra-226(n,2n)Ra-225 to Ac-225, with Ra-226(n,gamma)Ra-227 to Ac-227 as a competing branch. - Physics-informed LSTM, reduced five-nuclide system, 1,400 Radau-generated training scenarios, 6,000 epochs, seed 42. - Constant irradiation followed by exact decay-only cooling; activity reported per 1 g initial Ra-226. - Checkpoint and source hashes are preserved. No tuning is allowed after outside results are seen.	Post-lock reality checks - The LSTM beat a matched MLP on pointwise error, but the MLP ranked the full grid more accurately and ran faster. - Berkeley 33 and 40 MeV cases were outside training support; raw V3 activity predictions were about 600 times too high. - The reduced constant-rate equations have an exact analytic solution that is faster and more accurate than V3. - Current value is limited to screening candidates that are then checked with a trusted solver.
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Claim boundary: The project does not yet demonstrate reactor accuracy, experimental validation, a universal error below 3%, clinical readiness, production cost savings, or a decisive LSTM advantage.

One Deliverable Requested From Each Expert Route

Expert route	Requested technical deliverable	How it strengthens the study
Berkeley / LBNL	A reproducible 33/40 MeV specification: target-plane spectrum and normalization; target inventory, position, and geometry; beam/fluence history; irradiation, cooling, and separation times; produced versus recovered Ac-225 with uncertainty; any numerical Ac-227 detection limit.	Correct the published-case reconstruction and define a future model domain without tuning V3 to the answer.
NC State ARTISANS	A short critique of the frozen validation protocol, metrics, uncertainty, applicability-domain warning, and minimum evidence for a useful solver-verified surrogate.	Make the ML claim statistically and scientifically defensible.
MURR	A public example or realistic ranges for irradiation, cooling, separation, recovery, quality control, target recycling, and the constraints that dominate turnaround.	Replace an irradiation-only objective with an end-to-end workflow.
Johns Hopkins	A published reference for activity reference time, radionuclidic impurity quantity, uncertainty, and other outputs a production-screening tool should report.	Define a medically relevant objective without making a clinical claim.
UW-Madison	A public targetry/radiochemistry benchmark and guidance on beam, thermal, geometry, recovery, and turnaround variables for a realistic accelerator search.	Add feasibility limits and a stronger comparison problem.
MSU / FRIB	A public example for separation, recovery fraction, purification time, quality checks, and material losses between produced and usable isotope.	Add the missing chemistry and recovery layer to total-process time.
NIST	The recommended activity-to-atoms and reporting treatment for progeny equilibrium, assay/separation/reference times, decay data, and uncertainty.	Make experimental activity comparisons dimensionally and metrologically correct.

What I can provide Frozen checkpoint hash, model ranges, reduced reaction chain, full result tables, failure map, Berkeley reconstruction, source code, and assumption log.	What I will do with help Cite public sources, preserve negative results, document assumptions, verify finalists with a trusted solver, and acknowledge guidance only with permission.
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Supporting public sources: Morrell, *Next-Generation Isotope Production via Deuteron Breakup* (UC Berkeley, 2021); Berkeley Nuclear Data Program; NIST Ac-225 activity-standard work. All percentages above describe preserved computational artifacts, not facility or clinical performance.